

OPTI-CROP PROJECT DELIVERABLE

Problem-Solution Fit

1. Mapping Farmer Pains to Opti-Crop Features

The table below illustrates how the specific challenges faced by smallholder farmers are addressed by the design features of Opti-Crop.

Problem	Feature	Solution
Difficulty choosing crops due to unpredictable weather patterns.	Soil & Climate Predictor	Provides an accurate crop prediction based on current climate forecasts (rainfall, temp).
Soil degradation caused by blind application of urea/potash.	NPK Parameter Analysis	Analyzes specific nutrient shortages and tells the user exactly what fertilizer to add.
Complex, academic scientific reports from soil labs.	Actionable Advice Generator	Translates numeric parameters into simple, natural language advice (e.g. "Add lime").
Losing records of previous soil health evaluations.	User Search Logs Dashboard	Saves historical tests securely, enabling long-term tracking of soil recovery.

2. Product-Market Fit Validation

Opti-Crop targets a critical niche: bridging the gap between sophisticated data science (ML classifiers) and basic field farming. By making the app lightweight and local-first, it remains usable in low-connectivity environments where access to cloud computing is restricted, locking in a strong fit for rural agricultural markets.